TALS-CD: Causal Continuity-Based Task-Aware Mamba for Remote Sensing Binary Change Detection

Leiquan Wang, Lifa Xu, Chunbo Luo, and Chunlei Wu

***Abstract*—Scan strategies are critical for Mamba-based visual tasks, yet predefined handcrafted scan paths often fail when applied to task-specific data. In change detection (CD), irregular discontinuities between changed and unchanged regions disrupt spatial continuity, causing existing scan methods to increase effective dependency lengths and reduce the causal effectiveness of Mamba-based change feature representations. To address these challenges, a task-aware learnable scan (TALS) strategy is proposed for Mamba-based CD (TALS-CD). TALS learns causal correlations between local windows and establishes an interwindow priority sequence for bitemporal difference feature representation learning. With TALS, Mamba first models the global dependencies of different features and then progressively refines them in a top-down manner, leveraging the local properties of CNNs. In addition, a mixture of difference expert (MoDE) is introduced to utilize complementary difference representations, dynamically selecting optimal routing paths to enhance difference feature expression. Experimental results on three public datasets demonstrate that TALS-CD achieves superior CD performance with a small number of parameters and computational cost. The code is available at: https://github.com/fangcunxinghe/TALS-CD**

***Index Terms*—Change detection (CD), Mamba, remote sensing, state-space model (SSM), task-aware learnable scan (TALS).**

I. Introduction

REMOTE sensing change detection (CD) aims to identify land surface changes by comparing multitemporal imagery. As a core task in remote sensing and computer vision, CD plays a crucial role in applications like urban expansion monitoring [1], disaster damage assessment [2], land-use classification [3], and forest resource management [4]. Despite its importance, CD remains a challenging task due to factors such as complex scene variations, subtle-change patterns, the difficulty of accurately extracting change boundaries and edge-related features, and the challenge of achieving a balance between high detection accuracy and computational efficiency.

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CD has evolved from convolutional neural networks (CNNs) to Transformer-based models and more recently vision foundation model (VFM). CNN-based methods [9], [10] exploit local inductive biases and translation invariance to efficiently capture multilevel features, making them well suited for dense pixel-wise prediction. Transformer-based approaches [11], [12] further enhance detection performance by modeling long-range spatial–temporal dependencies through self-attention mechanisms. However, these performance gains often come at the cost of significantly increased model parameters, computational complexity, and memory consumption, leading to heavy training and inference overhead with diminishing accuracy improvements as model scale grows. Recently, VFM-based methods [13], [14], typically combined with parameter-efficient fine-tuning strategies, have shown strong generalization ability across datasets. Nevertheless, their reliance on large-scale pretrained backbones and high-resolution representations poses substantial challenges for deployment on resource-constrained platforms. Consequently, designing lightweight CD frameworks that effectively balance local detail modeling and global context aggregation while maintaining high detection accuracy remains an open and critical problem.

Mamba is an emerging and efficient sequence modeling architecture that has recently attracted significant attention in RSCD due to its linear time complexity and strong capability to capture long-range dependencies. The design of scanning strategies has become a crucial factor in enhancing the performance of Mamba-based models. Early approaches commonly adopted global Z-shaped [5], [6] or S-shaped [15] scan patterns, which, although intuitive, struggle to capture fine-grained local information. Subsequent improvements have attempted to alleviate this issue. LocalMamba [7] enables adaptive selection among predefined scanning directions, but still relies on fixed, manually designed scan templates. GrootV [16] constructs adaptive topologies based on local neighborhoods, enhancing local coherence but failing to capture long-range semantic dependencies. QuadMamba [17] adjusts window sizes according to regional information density, yet employs a fixed interwindow scanning order.

These Mamba-based variants primarily aim to improve spatial continuity in the scanning order, as illustrated in Fig. 1, but pay limited attention to causal continuity during sequence modeling. While spatial continuity supports local structure

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modeling, causal continuity determines how discriminative information is progressively aggregated along the sequence, which is particularly critical for CD. In the CD domain, CD-Lamba [18] prioritizes salient windows but overlooks the relational and causal dependencies among the remaining regions. Although DefMamba [19] attempts to learn the scanning order via index offsets, its optimization is hindered by long gradient propagation paths and sorting-induced gradient approximation, limiting effective modeling of causal dependencies along the sequence. Therefore, this motivates us to design a task-aware scanning mechanism that explicitly models causal continuity for CD.

Moreover, although significant progress has been made in CD using deep learning methods, this progress is primarily driven by deeper single-temporal semantic representations. However, bitemporal change difference representations remain underexplored, limiting performance in complex scenes and subtle-change scenarios. For example, some methods [20], [21], [22], [23] utilize feature subtraction for difference representation, others [24], [25], [26], [27] rely on concatenating channels to represent differences, and some [28], [29], [30], [31], [32] employ spatial attention mechanisms for difference representation. Despite these advancements, these basic difference representation approaches fail to meet the demands of CD in complex and diverse scenarios. It is difficult to design a single difference feature representation that can handle all types of changes. Different features capture complementary change cues from different perspectives. Therefore, leveraging their complementarity can help construct a more robust and discriminative bitemporal representation for CD.

To address the aforementioned challenges, we propose TALS-CD, a lightweight Mamba-based framework that specifically emphasizes the learning of bitemporal difference feature representations. The proposed framework adopts a coarse-to-fine strategy for CD. Specifically, a dual-stream bottom-up CNN–Mamba backbone extracts multiscale features from each image, where CNN blocks focus on fine-grained spatial cues and Mamba modules encode long-range semantic dependencies. To model high-level temporal differences, deep global features are fed into a task-aware difference enhancement (TADE) module, which leverages a task-aware learnable scan (TALS) strategy to construct scan paths by virtue of causal continuity for the CD task. This enables the network to generate a coarse top-down guidance map, analogous to human gist perception, which localizes potential change regions. Guided by this map, a mixture of difference expert (MoDE) modules refines pixel-, channel-, and spatial-level differences through a routing mechanism, progressively enhancing discriminative patterns. Finally, the refined features are fused and passed through a classifier to generate the final change map. In summary, the main contributions of this work are given as follows.

1) We reformulate the 2-D-to-1-D serialization in vision Mamba as a task-conditioned discrete interwindow scheduling problem. By predicting region-wise scan priorities from different features and deriving an explicit permutation to control the causal execution order of state-space model (SSM), the proposed TALS strategy enhances task-aligned causal continuity while preserving local coherence within each window.
2) We propose a TADE module that couples fixed-order bidirectional SSM difference modeling with task-adaptive TALS ordering, where task-aware cues generated under adaptive schedules modulate baseline global difference features to produce stronger global difference guidance.
3) We propose an MoDE with a learnable router to capture complementary difference cues. The router adaptively selects and fuses expert outputs across layers, enabling cross-layer difference refinement for CD.
4) Extensive experiments on public datasets, including LEVIR-CD, WHU-CD, and cropland CD (CLCD), demonstrate that the proposed TALS-CD achieves comparable performance to advanced methods while maintaining a lightweight network with reduced parameters, FLOPs, and GPU memory usage.

The rest of this article is organized as follows. Section II reviews the related work, Section III presents the proposed TALS-CD framework, and Section IV reports the experimental results. Finally, Section V concludes the article and outlines future research directions.

II. Related Work

A. CNN-Based CD Methods

CNN-based methods are among the earliest and most widely adopted deep learning frameworks for RSCD. Early convolutional approaches [9], [33] typically fuse bitemporal remote sensing images at the input stage, for example, via image concatenation, difference computation, or dimensionality reduction, so as to map multitemporal information into a single input representation. Classical semantic segmentation networks, such as U-Net [34] and FCN [35], are then employed to perform pixel-wise change prediction. Although these early fusion strategies are simple and straightforward, their detection performance is highly dependent on the design of the input fusion scheme, which limits their ability to model complex temporal variations.

To address this issue, siamese network architectures have been introduced, where bitemporal images are processed by shared or independent encoders to extract temporal features, followed by feature fusion at different hierarchical levels. This design explicitly models temporal differences across multiple spatial scales. Building upon this framework, numerous studies [10], [36], [37] further incorporate encoder–decoder architectures and multiscale feature fusion strategies, enabling end-to-end learning for pixel-level CD. By effectively integrating low-level spatial details with high-level semantic information, these methods achieve improved performance in delineating change boundaries and detecting small objects. In addition, various channel and spatial attention mechanisms have been introduced to suppress background noise and emphasize change-relevant regions.

Despite the notable progress achieved by CNN-based methods, their core operations remain inherently local, resulting in limited receptive fields. Although deeper architectures or

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dilated convolutions can partially alleviate this limitation, they are still insufficient for modeling long-range dependencies and global contextual relationships. Consequently, in complex urban scenes and natural environments, CNN-based methods often struggle to capture large-scale structural changes or globally correlated change patterns.

B. Transformer-Based CD Methods

To overcome the limitations of local receptive fields in CNNs, Transformer-based methods have been introduced into RSCD by leveraging self-attention mechanisms for global dependency modeling. In recent years, vision Transformers and their variants have been widely explored in RSCD [12], [38], [39], [40], [41]. These approaches enable long-range interaction modeling over the entire feature map, thereby improving the representation of spatially distant regions and enhancing robustness under complex background conditions.

Some studies directly construct CD models based on Transformer architectures. For example, Zhang et al. [42] adopt a siamese U-shaped framework with multiple Swin Transformer blocks for feature encoding, fusion, and decoding of bitemporal images, enabling multiscale global context modeling. Other works integrate Transformers with CNN backbones to balance local feature extraction and global contextual modeling. In such hybrid frameworks, convolutional layers capture fine-grained spatial details, while Transformer modules model global relationships between temporal features. Chen et al. [11] extract high-level semantic features from bitemporal images using CNNs and subsequently employ a Transformer encoder–decoder to enhance temporal feature representation, resulting in improved CD performance.

Despite the performance gains achieved by these hybrid architectures, the core component of Transformers—the self-attention mechanism—suffers from inherent computational bottlenecks, as its computational complexity scales quadratically with the number of input tokens. With increasing image resolution and feature map size in remote sensing scenarios, the number of tokens grows rapidly, resulting in substantial memory consumption and inference latency. This significantly limits the efficiency and scalability of Transformer-based methods for high-resolution RSCD tasks.

C. VFM-Based CD Methods

Recently, VFM-based methods [43] have explored large-scale pretrained VFM to further enhance the generalization capability of RSCD systems. By fine-tuning or adapting foundation models pretrained on massive vision–language or image datasets, these approaches transfer rich semantic priors to remote sensing tasks, achieving promising performance across different geographical regions and spatial resolutions.

In particular, some works [44] leverage vision–language models to align bitemporal images with semantic prompts, enabling zero-shot or few-shot CD. Other studies [13], [14] incorporate foundation models as feature extractors or auxiliary supervision modules to improve representation robustness. Despite their strong generalization ability, VFM-based methods usually involve heavy computational resources and complex training pipelines. The large model size and high inference cost remain major obstacles to efficient and scalable deployment in practical remote sensing applications.

D. SSM-Based CD Methods

In recent years, SSM-based methods [18], [45], [46] have been introduced into RSCD as an efficient alternative to Transformer-based architectures. By reformulating global dependency modeling as a sequence modeling problem with linear computational complexity, the Mamba architecture enables long-range information propagation while effectively reducing computational and memory costs. Chen et al. [47] first introduced Mamba-based state-space modeling for efficient global context modeling in RSCD, while Zhang et al. [48] further enhanced local spatial feature representation through convolutional integration.

Despite these advances, most existing SSM-based RSCD methods still rely on fixed scanning paths and directly flatten 2-D feature maps into 1-D sequences for modeling. Such designs inevitably disrupt the inherent spatial continuity of remote sensing imagery and assign equal modeling priority to all regions, treating change-relevant areas and unchanged background regions in the same manner. This limitation restricts both modeling efficiency and fine-grained change discrimination capability.

In contrast, this article proposes a task-aware dynamic scanning strategy that adaptively assigns higher modeling priority to change-relevant regions. By guiding the scanning order in a task-driven manner, the proposed method enhances spatial dependency modeling while preserving the efficiency advantages of SSM-based architectures, achieving a more favorable tradeoff between model complexity and detection performance.

III. Methodology

A. Overall Architecture

To address the limitations of fixed scan patterns in CD, this work introduces learn to scan: TALS, a framework that enables the model to learn task-aware scan paths for CD. The core idea follows a coarse-to-fine, global-to-local perception paradigm, inspired by human visual cognition: the model first forms a global gist to localize potential change regions, and then progressively refines local details under top-down guidance, ultimately resolving pixel-level differences.

Fig. 2 illustrates the overall architecture, which is implemented in three phases.

1) *Single-Temporal Feature Extraction:* A bottom-up CNN–Mamba backbone for single-temporal feature extraction that integrates local fine-grained features from CNNs and global semantic dependencies via Mamba's long-sequence modeling.
2) *Global Difference Modeling:* A TADE module employs a TALS strategy in Mamba to modulate and enhance

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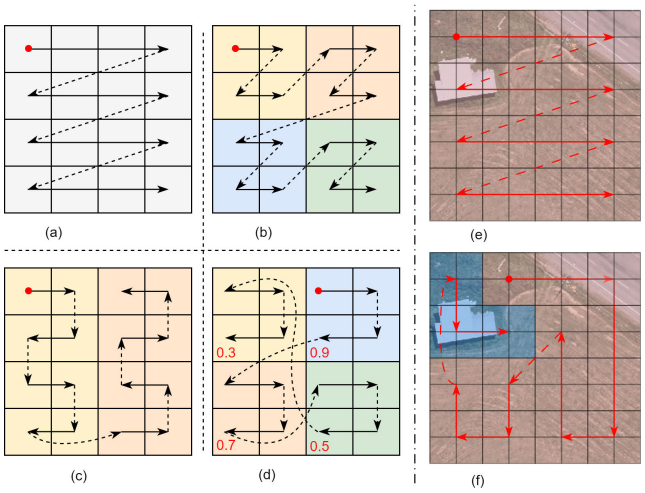


Fig. 1. Comparison of scanning strategies in Mamba-based vision models and our approach. (a) ViM [5]/V-Mamba [6] uses Z-shaped flattening from 2-D to 1-D, breaking spatial locality. (b) LocalMamba's [7] local scans keep locality, but Z-paths within/across windows break spatial continuity. (c) MaIR's [8] nonoverlapping stripes with S-scans preserve locality and spatial continuity, yet human-crafted spatial continuity does not imply causal continuity, which may not be suitable for specific tasks like CD (e.g., BCD). (d) TALS learns the start point and scan path automatically to achieve task-aware causal continuity. (e) Spatial continuity prioritizes local adjacency in the scanning order. (f) Causal continuity emphasizes consistent reasoning over change-related regions.

global difference representations, providing top-down guidance.

3) *Local Difference Refinement:* An MoDE with a routing mechanism that performs hierarchical refinement across pixel-, channel-, and spatial-level differences. A classifier aggregates the refined features to generate the final change map. The network is trained with a cross-entropy loss and is designed to remain computationally efficient, maintaining global causal continuity and local spatial coherence throughout the pipeline.

In this pipeline, TALS, TADE, and MoDE are tightly coupled. Specifically, TALS is embedded within TADE to learn task-aware scan priorities and enhance global difference representations, whose resulting activation maps provide coarse guidance for subsequent refinement. Guided by these task-aware cues, MoDE further performs adaptive fusion and refinement of multilevel difference features. Through this interaction, global difference modeling and local refinement are effectively coordinated in a coarse-to-fine manner.

B. Task-Aware Learnable Scan Strategy

In this section, we introduce a TALS strategy for Mamba, aiming to overcome the limitations of fixed scan patterns by explicitly aligning the scanning order with task semantics. The core idea of TALS is to jointly preserve spatial continuity and enhance causal continuity by learning task-specific scan priorities.

As illustrated in Fig. 3, given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, we first aim to estimate region-level importance while retaining sufficient spatial context. To this end, a sequence of convolutional layers with kernel sizes 7×7, 5×5, and 3×3 is applied to progressively enlarge the receptive field and aggregate multiscale contextual information

$$X^{(3)} = \mathrm{Conv}_{3\times 3}\left(\mathrm{Conv}_{5\times 5}\left(\mathrm{Conv}_{7\times 7}(X)\right)\right). \tag{1}$$

By progressively expanding the receptive field, each spatial location encodes broader spatial dependencies, which are crucial for reliably assessing the importance of local regions in CD. Based on $X^{(3)}$, we then perform a nonoverlapping 2-D convolution with a kernel size $K\times K$ and a stride K, producing one response per $K\times K$ windows. This design enforces local spatial aggregation and naturally preserves intrawindow spatial continuity. A sigmoid activation is applied to obtain the task-aware activation map

$$X_{\mathrm{act}} = \sigma\big(\mathrm{Conv}(X^{(3)})\big) \in \mathbb{R}^{1\times \frac{H}{K}\times \frac{W}{K}}. \tag{2}$$

Each entry $X_{\mathrm{act}}(i,j) \in (0,1)$ reflects the relative priority of the corresponding $K\times K$ regions in X with respect to the current task, serving as a task-aware prior that guides the subsequent scanning order.

To enhance causal continuity across spatial regions, the activation map X_{act} is flattened and sorted in descending order, yielding a permutation π that defines the interwindow scanning sequence. Windows with higher task relevance are therefore processed earlier, allowing discriminative information to be propagated along the sequence in a task-aligned causal manner. The input feature map X is serialized according to π and fed into the SSM for sequential modeling. Within each window, a standard S-shaped scan path is adopted to preserve local spatial coherence. This hierarchical scanning strategy ensures global, priority-aligned causal continuity across windows while maintaining spatial continuity within each window. After sequence modeling, the inverse permutation π^{-1} is applied to restore the features to their original spatial layout.

Although the sorting operation is implemented as a hard permutation in the forward pass, TALS remains learnable during training. Following [19], a gradient approximation strategy is employed in the backward pass to enable gradient propagation through the sorting operation, allowing scan priorities to be optimized in an end-to-end manner.

However, due to the long-sequence nature of Mamba, gradients propagated from the sequence tail to the head can be weak and unstable, making it difficult to learn a reliable and semantically meaningful scan priority solely through indirect supervision. To alleviate this issue, we introduce an auxiliary binary cross-entropy (BCE) loss on the task-aware activation map. This auxiliary supervision provides direct, coarse-grained task guidance, encouraging the activation map to highlight change-relevant regions and stabilizing the learning of scan priorities. As a result, TALS enables more effective and robust modeling of task-aligned causal continuity.

C. TADE Module

To address the limitations of purely local difference modeling and fixed scanning strategies in CD, we propose a *TADE* module built upon the Mamba architecture. TADE aims to explicitly model global difference dependencies while introducing task-aware causal guidance in a stable and effective manner.

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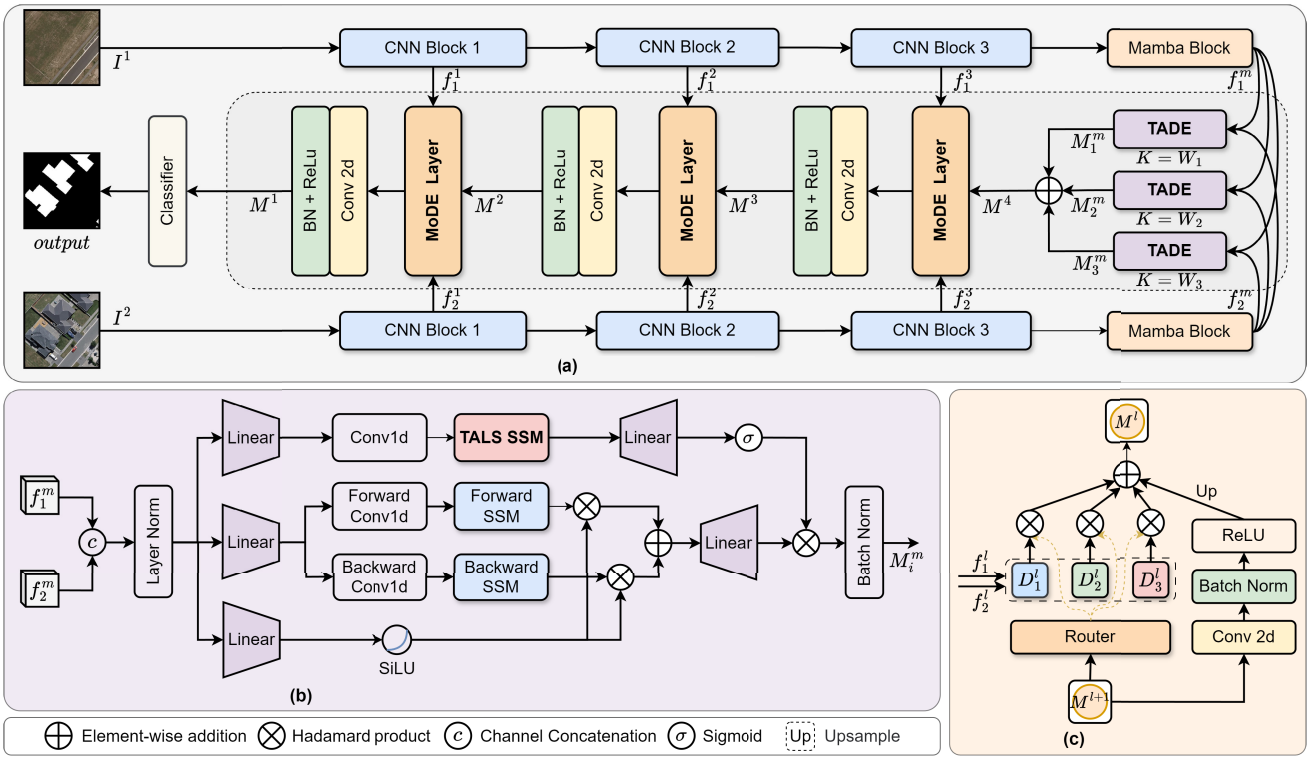


Fig. 2. Overall architecture. The architecture follows a gist-to-detail paradigm aligned with human visual cognition: global differences are first extracted to propose candidate change regions, after which top–down guidance drives progressively finer refinements across scales, ultimately resolving pixel-level differences. Concretely, single-temporal features are extracted by a dual-stream CNN–Mamba backbone, the TADE module enhances a global difference map through learnable scan paths, and the MoDE module then progressively refines pixel-, channel-, and spatial-level cues for final change prediction. (a) Overall architecture. (b) TADE. (c) MoDE layer.

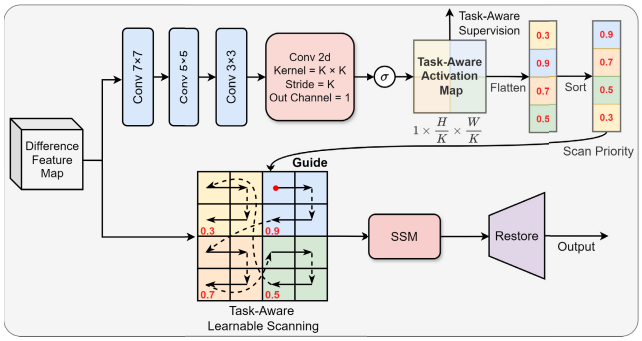


Fig. 3. Task-aware learnable scan strategy. The strategy learns region-wise scan priorities from different features and guides the sequential processing order of Mamba to enhance task-aligned causal modeling.

As illustrated in Fig. 2(b), the bitemporal features f_1^m and f_2^m are first concatenated and normalized, and then fed into three parallel branches with complementary functionalities. 1) *Main Branch:* For stable global difference modeling under a fixed scan order, prioritizing robust spatial dependency modeling in the primary data flow. 2) *TALS-Enhanced Branch:* For learning task-aware causal scan priorities, explicitly modeling causal continuity aligned with the CD objective. 3) *Auxiliary Modulation Branch:* For adaptive feature reweighting, this hybrid architecture facilitates the coordinated modeling of spatial continuity and task-aware causal continuity.

In the main branch, the input features are first projected through a linear layer, followed by a 1-D convolution and an SSM. Bidirectional scanning is performed along the sequence dimension to capture global contextual dependencies in both forward and backward directions. By adopting a fixed scan order, this branch ensures stable spatial modeling and reliable training behavior, producing the baseline global difference representations.

In the auxiliary modulation branch, attention weights are generated via a linear projection followed by a SiLU activation. These weights are applied through element-wise multiplication to independently modulate the forward and backward features from the main branch. The modulated features are then aggregated by summation and projected through a linear layer, yielding an adaptively weighted global difference representation.

In parallel, the TALS-enhanced branch performs task-aware global difference modeling by dynamically learning scan priorities that emphasize change-relevant regions. Unlike the main branch, this pathway focuses on modeling causal dependencies aligned with the CD objective, without disrupting the stable global modeling process. Specifically, the features are processed sequentially by a linear projection and a 1-D convolution, and then fed into an SSM equipped with the TALS strategy. The resulting task-oriented global difference features are further transformed by a linear layer followed by a sigmoid activation to generate task-aware modulation cues.

Finally, the baseline global difference features from the main branch, denoted as $\mathcal{F}_{\text{main}}$, are modulated by the task-aware cues from the TALS-enhanced branch, denoted as $\mathcal{F}_{\text{TALS}}$,

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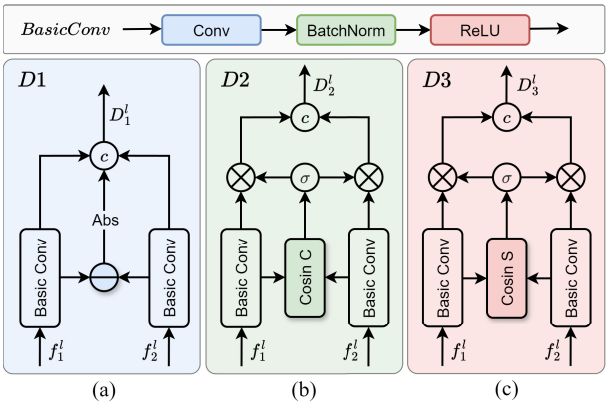


Fig. 4. Three CNN-based difference experts. (a) Pixel-wise absolute-difference expert. (b) Channel-similarity-guided difference expert. (c) Spatial-similarity-guided difference expert.

through element-wise multiplication. The fused features are then normalized using batch normalization to obtain the global difference map M^m

$$M^m = \mathrm{BN}\big(\mathcal{F}_{\mathrm{main}} \odot \sigma(\mathcal{F}_{\mathrm{TALS}})\big) \tag{3}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function.

Although TALS enables task-adaptive scan paths for CD, its modeling capacity remains constrained by the local window size K. To alleviate this limitation, multiple TADE modules with different window sizes are employed to exploit multiscale complementarity. This design allows the model to capture potential changes across diverse spatial extents and levels of granularity. Specifically, three TADE modules are constructed with window sizes set to W_1–W_3. The selection of these hyperparameters is discussed in Section IV.

D. Mixture of Difference Experts

To overcome the limitations of a single bitemporal difference cue, an MoDE module with a routing mechanism is introduced for adaptive fusion of diverse difference representations, thereby enabling top-down refinement across stages with varying resolutions.

1) Difference Experts: In Fig. 4, three complementary CNN-based difference experts are designed for RSCD: a pixel-wise absolute-difference expert D_1, a channel-similarity-guided difference expert D_2, and a spatial-similarity-guided difference expert D_3. Assume that at the single-temporal feature extraction stage, the lth layer produces CNN features $f_1^l, f_2^l \in \mathbb{R}^{c\times h\times w}$ for the two images, where c, h, and w denote the number of channels, height, and width, respectively. Each expert begins with a basic convolutional block (Conv–BN–ReLU), applied separately to each temporal stream

$$\hat{f}_t^l = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}(f_t^l))\big), \quad t \in \{1, 2\}. \tag{4}$$

To estimate cross-temporal similarity, channel-wise and spatial-wise similarity maps are defined as follows:

$$S_c = \sigma\big(\mathrm{CosC}\big(\hat{f}_1^l, \hat{f}_2^l\big)\big) \tag{5}$$

$$S_s = \sigma\big(\mathrm{CosS}\big(\hat{f}_1^l, \hat{f}_2^l\big)\big) \tag{6}$$

where $\sigma(\cdot)$ denotes the sigmoid function. CosC$(\cdot)$ computes the cosine similarity across channels at each spatial location and returns $S_c \in \mathbb{R}^{1\times h\times w}$. CosS$(\cdot)$ computes the cosine similarity across the spatial grid for each channel and returns $S_s \in \mathbb{R}^{c\times 1\times 1}$.

Consequently, the difference maps produced by the pixel-wise absolute-difference expert D_1, the channel-similarity-guided expert D_2, and the spatial-similarity-guided expert D_3 at layer l are computed as follows:

$$D_1^l = \mathrm{Concat}\big(\hat{f}_1^l, \hat{f}_2^l, |\hat{f}_1^l - \hat{f}_2^l|\big) \tag{7}$$

$$D_2^l = \mathrm{Concat}\big(S_c \odot \hat{f}_1^l, S_c \odot \hat{f}_2^l\big) \tag{8}$$

$$D_3^l = \mathrm{Concat}\big(S_s \odot \hat{f}_1^l, S_s \odot \hat{f}_2^l\big) \tag{9}$$

where Concat$(\cdot)$ denotes the channel-wise concatenation and $\odot$ denotes the element-wise product used to apply the similarity maps to the corresponding features.

2) Routing Mechanism: MoDE employs a routing mechanism to adaptively integrate multiexpert difference cues, as shown in Fig. 2(d). Instead of concatenating or averaging features, the router gates the contributions of the three experts at layer l using context from the $(l+1)$th layer.

Given the difference map from the upper layer M^{l+1}, global average pooling extracts context, which is passed through two linear layers with the ReLU nonlinearity and then a sigmoid to produce the expert weights

$$\big[\omega_1^l, \omega_2^l, \omega_3^l\big] = \sigma\big(\mathrm{Linear}\big(\mathrm{ReLU}(\mathrm{Linear}\big(\mathrm{GAP}(M^{l+1})\big))\big)\big) \tag{10}$$

where GAP$(\cdot)$ denotes global average pooling and $\sigma(\cdot)$ is the sigmoid used for normalization. The three experts are then fused according to their weights

$$D^l = \sum_{i=1}^{3} \omega_i^l D_i^l. \tag{11}$$

For cross-layer alignment, the upper layer feature is first transformed by the convolution, the batch normalization, and the ReLU activation. After upsampling, the refined representation at layer l is obtained by residual fusion

$$M^l = D^l + \mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}(M^{l+1})\big)\big) \tag{12}$$

where Up$(\cdot)$ denotes the upsampling operation. Leveraging the routing mechanism of MoDE, TALS-CD achieves top-down, progressive refinement of difference features across layers.

E. Loss Function

The binary CD (BCD) task is formulated as a pixel-wise binary classification problem. During training, supervision is applied not only to the final prediction but also to the task-aware activation maps generated by the TALS scanning mechanism, enabling effective guidance for learning task-adaptive scan paths.

1) Main Loss: Let P denote the final predicted change probability map produced by the network, and $Y \in \{0, 1\}^{H\times W}$ be the corresponding ground-truth change label. The main supervision is imposed using the BCE loss

$$\mathcal{L}_{\mathrm{main}} = -\frac{1}{N}\sum_{i=1}^{N}\big[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\big] \tag{13}$$

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where N denotes the total number of pixels. p_i and y_i represent the predicted probability and ground-truth label of the ith pixel, respectively.

2) Auxiliary Task-Aware Loss: To explicitly supervise the task-aware scanning behavior, an auxiliary loss is applied to the task-aware activation maps generated by the TALS-enhanced branches. Specifically, three TADE modules with different window sizes are employed to capture multiscale change patterns, each producing a task-aware activation map $\{A^{(s)}\}_{s=1}^{3}$. Since the activation maps are generated at different spatial resolutions, each $A^{(s)}$ is first resized to the input resolution to align with the ground-truth label Y. For each scale, an auxiliary BCE loss is then defined as follows:

$$\mathcal{L}\text{aux}^{(s)} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(a_i^{(s)}) + (1-y_i)\log(1-a_i^{(s)})\right] \tag{14}$$

where $a_i^{(s)}$ denotes the activation value at pixel i from the resized activation map of the sth TADE module. These auxiliary losses encourage the task-aware activation maps to highlight regions corresponding to true changes, thereby facilitating the learning of discriminative scan paths.

Without such supervision, the scan priorities in TALS would be implicitly learned only through the final detection objective, which provides indirect and weak signals for spatial prioritization. This may lead to suboptimal scan ordering and unstable optimization. By introducing task-aware supervision, the model receives explicit guidance to emphasize task-relevant regions and suppress background noise, enabling more effective and interpretable learning of scan priorities.

3) Overall Loss: The overall training objective is formulated as a weighted combination of the main loss and the auxiliary task-aware losses

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \alpha \sum_{s=1}^{3} \mathcal{L}_{\text{aux}}^{(s)} \tag{15}$$

where α is a weighting factor balancing the contribution of auxiliary supervision. Moreover, the proposed task-aware supervision is not limited to CD and can be extended to other dense prediction tasks by designing task-specific supervision signals for spatial prioritization.

IV. Experimental Results and Analysis

A. Datasets

1) WHU-CD [33]: The WHU-CD dataset, a subset of the larger WHU Building dataset, is specifically designed for building CD. It consists of a pair of aerial images of Christchurch, New Zealand, captured in 2012 and 2016 with a spatial resolution of 0.3 m. The dataset reflects significant urban expansion during this four-year period. According to the official partitioning scheme, 5948, 743, and 743 image pairs are used for training, validation, and testing, respectively.

2) LEVIR-CD [49]: The LEVIR-CD dataset contains 637 pairs of bitemporal remote sensing images collected from Google Earth, each with a size of 1024 × 1024 pixels and a spatial resolution of 0.5 m. It includes over 31 000 manually annotated building change instances from various urban areas. Following the standard evaluation protocol, we employ 445 pairs for training, 64 for validation, and 128 for testing.

3) CLCD [50]: The CLCD dataset targets land-cover CD in agricultural regions. It comprises 600 pairs of bitemporal images acquired by the Gaofen-2 satellite over Guangdong Province, China, between 2017 and 2019. Each image has a size of 512 × 512 pixels, with spatial resolution ranging from 0.5 to 2 m. The dataset encompasses diverse change categories such as buildings, roads, water bodies, and bare land. In accordance with the official guideline, we divide the dataset into 360 pairs for training, 120 for validation, and 120 for testing.

B. Experimental Setup

1) Implementation Details: All experiments are implemented in PyTorch and conducted on an RTX 4090 GPU with 24 GB of memory. For the LEVIR-CD dataset, we train the model for 1000 epochs with an input size of 512 × 512 and a batch size of 16. For the WHU-CD dataset, we train for 100 epochs using the original input size of 256 × 256 and a batch size of 32. For the CLCD dataset, the model is trained for 500 epochs with a crop size of 512 × 512 and a batch size of 16. The number of training epochs is adjusted according to the dataset size to ensure a comparable number of training iterations across datasets. We adopt the AdamW optimizer with an initial learning rate of 0.001, $\beta = (0.9, 0.999)$, and a weight decay of 0.05. A linear warm-up strategy is applied during the first 15 epochs to gradually increase the learning rate from 0 to the set value, followed by a cosine annealing schedule for the remaining epochs. To enhance generalization, data augmentation techniques are applied, including random cropping, flipping, rotation, and adjustments of brightness and contrast.

2) Evaluation Metrics: We evaluate our BCD models using standard pixel-level metrics, including precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and overall accuracy (OA). We adopt $F1$ and IoU as the primary evaluation metrics, while the others provide complementary assessment.

C. Performance Comparisons

To evaluate the effectiveness of our method on BCD tasks, we compare the proposed TALS-CD with a comprehensive set of benchmark approaches covering four main categories: 1) *CNN-based methods:* including IFNet [36] and SNUNet [10]; 2) *Transformer-based methods:* such as BIT [11], ChangeFormer [12], ICIF-Net [51], DMINet [41], AMTNet [38], EATDer [39], SSCD [40], PHENet [52], and FarmCD [53]; 3) *Mamba-based methods:* including RS-Mamba [45], ChangeMamba [47], SPMNet [24], CDMamba [48], CDxLSTM [46], and CD-Lamba [18]; and 4) *VFM-based methods:* such as ChangeCLIP [44], BAN [13], SAM-CD [14], and SFCD-Net [43].

1) Quantitative Results: Quantitative evaluation is conducted on three widely used benchmarks, with comparisons against state-of-the-art methods based on CNNs, Transformers, and Mamba in Table I. A separate comparison is provided

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TABLE I

QUANTITATIVE RESULTS OF CNN-, TRANSFORMER-, AND MAMBA-BASED MODELS ON THREE BENCHMARK DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, AND ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Type	Models	#Params (M)	FLOPs (G)	WHU-CD Pre. / Rec. / F1 / IoU / OA	LEVIR-CD Pre. / Rec. / F1 / IoU / OA	CLCD Pre. / Rec. / F1 / IoU / OA
C	$IFNet_{20}$ [36]	50.71	41.18	96.91 / 73.19 / 83.40 / 71.52 / 98.83	94.02 / 82.93 / 88.13 / 78.77 / 98.87	59.08 / 66.39 / 62.52 / 45.48 / 94.08
	$SNUNet_{21}$ [10]	12.04	54.82	85.60 / 81.49 / 83.50 / 71.67 / 98.71	89.18 / 87.17 / 88.16 / 78.83 / 98.82	71.42 / 53.78 / 61.36 / 44.25 / 94.96
T	BIT_{22} [11]	3.55	10.63	86.64 / 81.48 / 83.98 / 72.39 / 98.75	89.24 / 89.37 / 89.31 / 80.68 / 98.92	72.47 / 66.93 / 69.59 / 53.36 / 95.64
	$ChangeFormer_{22}$ [12]	41.03	202.79	93.73 / 87.11 / 90.30 / 82.32 / 99.26	92.05 / 88.80 / 90.40 / 82.48 / 99.04	72.37 / 63.12 / 67.43 / 50.87 / 95.46
	$ICIF\text{-}Net_{22}$ [51]	23.82	25.36	92.98 / 85.56 / 88.32 / 79.24 / 98.96	91.32 / 88.64 / 89.96 / 81.75 / 98.99	72.14 / 65.58 / 68.66 / 52.27 / 95.77
	$DMINet_{23}$ [41]	6.24	14.42	93.84 / 86.25 / 88.69 / 79.68 / 98.97	92.52 / 89.95 / 90.71 / 82.99 / 99.07	72.37 / 62.97 / 67.24 / 50.65 / 95.21
	$AMTNet_{23}$ [38]	24.67	21.56	92.86 / 91.99 / 92.27 / 85.64 / 99.32	91.82 / 89.71 / 90.76 / 83.08 / 98.96	78.64 / 75.06 / 76.81 / 62.35 / 96.55
	$EATDer_{24}$ [39]	6.61	23.43	91.32 / 88.74 / 90.01 / 81.97 / 98.58	91.74 / 90.67 / 91.20 / 83.80 / 98.75	74.55 / 69.53 / 72.01 / 56.19 / 96.11
	$SSCD_{24}$ [40]	3.22	9.26	92.45 / 88.74 / 90.19 / 82.13 / 99.27	92.12 / 89.08 / 90.58 / 82.78 / 99.06	76.19 / 65.01 / 70.16 / 54.03 / 95.88
	$PHENet_{26}$ [52]	5.24	77.50	93.56 / 90.91 / 92.22 / 85.56 / 99.33	92.29 / 89.41 / 90.83 / 83.20 / 99.06	80.11 / 65.79 / 72.25 / 56.55 / 96.23
	$FarmCD_{26}$ [53]	15.95	48.02	-	-	78.54 / 69.31 / 73.63 / 58.26 / 96.19
M	$RS\text{-}Mamba_{24}$ [45]	51.95	22.82	93.37 / 90.42 / 91.87 / 84.96 / 99.33	82.52 / 89.73 / 91.10 / 83.66 / 99.10	72.18 / 66.77 / 69.37 / 53.11 / 95.61
	$ChangeMamba\text{-}T_{24}$ [47]	17.13	45.74	95.02 / 89.84 / 92.36 / 85.80 / 99.35	92.09 / 90.22 / 91.14 / 83.73 / 99.11	80.80 / 73.36 / 76.90 / 62.47 / 96.72
	$SPMNet_{25}$ [24]	17.33	6.94	94.67 / 89.10 / 91.80 / 84.84 / 99.31	92.12 / 90.58 / 90.99 / 84.07 / 99.08	-
	$CDMamba_{25}$ [48]	11.90	54.18	95.58 / 92.01 / 93.76 / 88.26 / 99.51	91.43 / 90.08 / 90.75 / 83.07 / 99.06	74.78 / 69.05 / 71.81 / 56.02 / 95.96
	$CDxLSTM_{25}$ [46]	16.19	3.92	93.71 / 91.49 / 92.58 / 86.19 / 99.33	91.52 / 90.27 / 90.89 / 83.30 / 99.07	83.15 / 74.76 / 78.73 / 64.92 / 96.99
	$CD\text{-}Lamba_{25}$ [18]	28.74	15.26	93.45 / 91.59 / 92.51 / 86.07 / 99.32	-	79.20 / 76.96 / 78.06 / 64.02 / 96.78
	TALS-CD (Ours)	1.90	1.84	95.99 / 91.77 / 93.83 / 88.38 / 99.47	92.42 / 91.00 / 91.71 / 84.68 / 99.16	82.72 / 73.44 / 77.80 / 63.67 / 96.88

TABLE II

QUANTITATIVE RESULTS OF VFM-BASED MODELS ON THREE BENCHMARK DATASETS. ALL VALUES ARE REPORTED IN PERCENTAGE (%). "*" DENOTES THE COUNT OF TRAINABLE PARAMETERS ONLY

Type	Models	#Params (M)	FLOPs (G)	WHU-CD Pre. / Rec. / F1 / IoU / OA	LEVIR-CD Pre. / Rec. / F1 / IoU / OA	CLCD Pre. / Rec. / F1 / IoU / OA
V	$ChangeCLIP_{24}$ [44]	117.54	130.44	95.63 / 94.02 / 94.82 / 90.15 / 99.52	93.40 / 90.67 / 92.01 / 85.20 / 99.20	-
	BAN_{24} [13]	3.80*	330.32	94.55 / 93.73 / 93.99 / 88.68 / 99.48	92.83 / 90.89 / 91.85 / 84.93 / 99.12	-
	$SAM\text{-}CD_{24}$ [14]	-	-	96.67 / 93.64 / 95.13 / 90.71 / 99.62	93.58 / 90.93 / 92.24 / 85.59 / 99.22	83.17 / 71.60 / 76.95 / 62.54 / 96.81
	$SFCD\text{-}Net_{24}$ [43]	14.56*	273.43	95.50 / 94.95 / 95.23 / 90.89 / 99.63	92.79 / 92.40 / 92.59 / 86.19 / 99.21	-
	TALS-CD (Ours)	1.90	1.84	95.99 / 91.77 / 93.83 / 88.38 / 99.47	92.42 / 91.00 / 91.71 / 84.68 / 99.16	82.72 / 73.44 / 77.80 / 63.67 / 96.88

for VFM methods in Table II, considering their strong cross-domain generalization and scalability, but also their limitations in terms of high computational cost.

As shown in Table I, the proposed TALS-CD demonstrates competitive performance compared with existing approaches. Notably, our model consistently outperforms representative CNN- and Transformer-based methods: compared with SNUNet and PHENet, it improves $F1$ by 10.33%/3.55%/16.44% and 1.61%/0.88%/5.55% on the three datasets, respectively. When compared with Mamba-based approaches (CDMamba and CDxLSTM), our model further yields $F1$ gains of 0.07%/0.96% and 1.25%/0.82% on WHU-CD and LEVIR-CD, respectively. Although the $F1$ on CLCD is slightly lower than CDxLSTM and CD-Lamba by 0.93% and 0.26%, our model uses only 11.7% and 6.6% of their parameters, and requires only 46.9% and 12.0% of their computational cost, respectively. It is particularly noteworthy that under the 10M parameter setting, our method achieves $F1$ gains of 3.64%, 0.51%, and 5.81% over the second-best method on the three datasets, highlighting a favorable tradeoff between accuracy and efficiency.

As reported in Table II, VFM-based models achieve strong performance but at substantially higher parameter and computational budgets. In contrast, our model attains comparable accuracy while remaining far more compact: it is only 1.40% and 0.88% lower in $F1$ on WHU-CD and LEVIR-CD, respectively, yet requires a much smaller model size and computational load. This further supports the efficacy of our approach for practical and resource-constrained deployment scenarios.

2) Qualitative Results: Figs. 5–7 show qualitative visual comparisons on the WHU-CD, LEVIR-CD, and CLCD datasets, respectively. For fairness and brevity, we restrict the comparison to Transformer-based methods (BIT, ChangeFormer, SSCD, and PHENet) and Mamba-based methods (RS-Mamba, ChangeMamba, and CDMamba). CNN-based approaches are excluded due to their relatively lower accuracy, while VFM-based methods are omitted because of their substantially larger parameter scales.

On WHU-CD, the proposed TALS-CD delineates new building boundaries with clear, coherent edges, whereas others blur or fragment edges, and CDMamba produces large false positives in dense areas. On LEVIR-CD, a subtle building change is precisely detected, while competing methods miss the small-scale variation. On CLCD, despite lower resolution and cloud–haze–illumination interference, the proposed method suppresses both false positives and false negatives and preserves clean region boundaries.

The performance gains stem from two key factors: TADE, which introduces task-aware global guidance by using a TALS-enhanced branch to learn causal scan orders that reinforce fixed scan Mamba representations, and the MoDE router, which adaptively integrates pixel, channel, and spatial cues to refine change representations in a coarse-to-fine manner.

3) Model Efficiency: Efficiency is assessed in terms of parameter count, FLOPs, and GPU resource usage. Fig. 8

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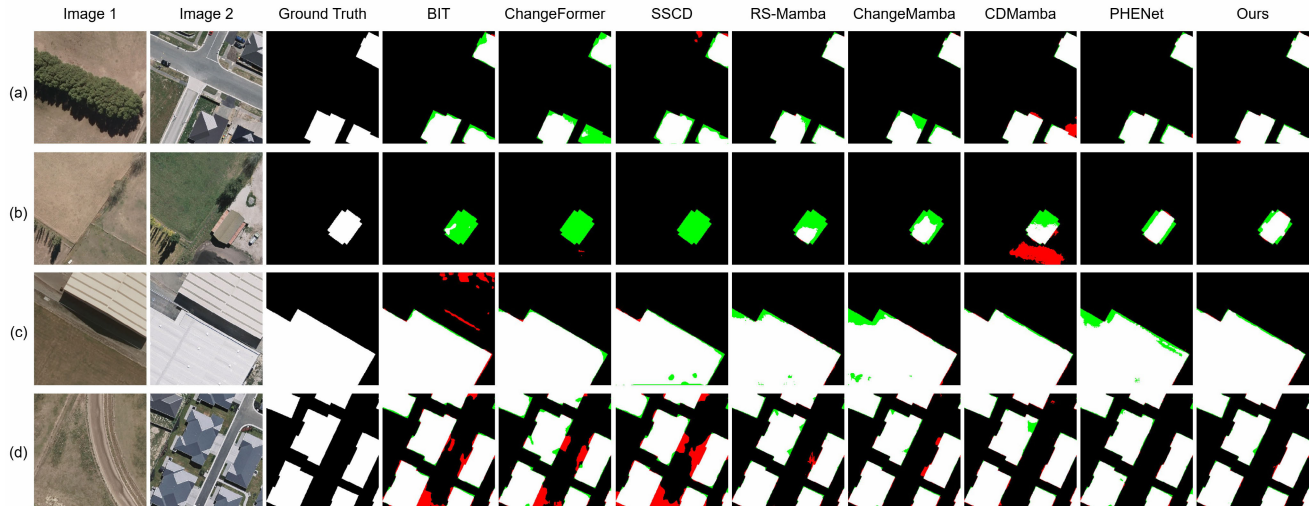


Fig. 5. Visualization results for qualitative comparison on the WHU-CD dataset. (a)–(d) Representative samples. Colors denote detection outcomes: true positives (white), true negatives (black), false positives (red), and false negatives (green).

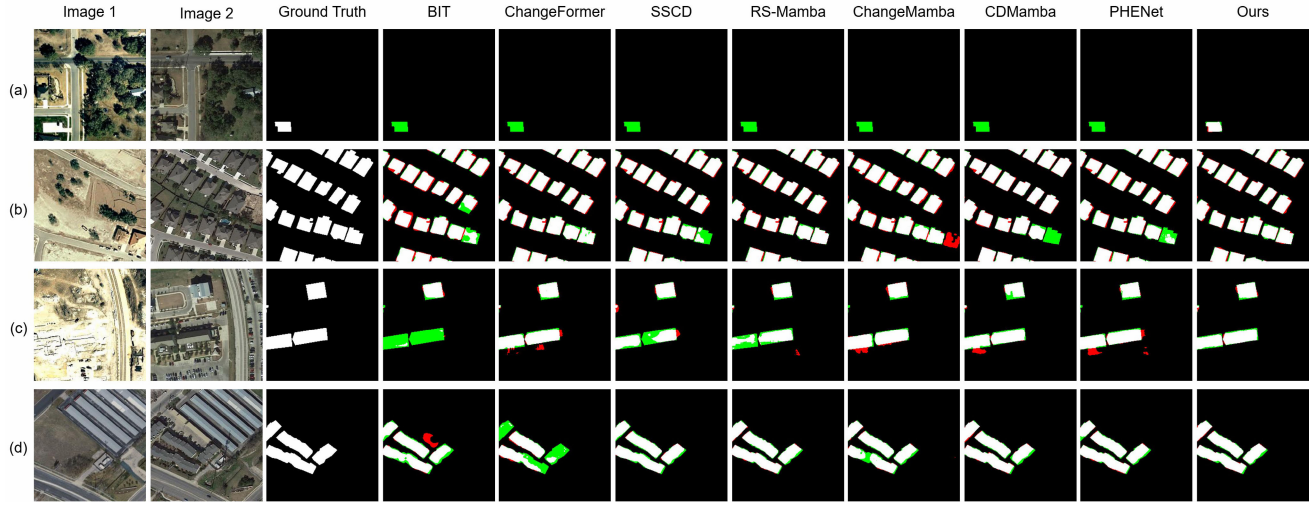


Fig. 6. Visualization results for qualitative comparison on the LEVIR-CD dataset. (a)–(d) Representative samples. Colors denote detection outcomes: true positives (white), true negatives (black), false positives (red), and false negatives (green).

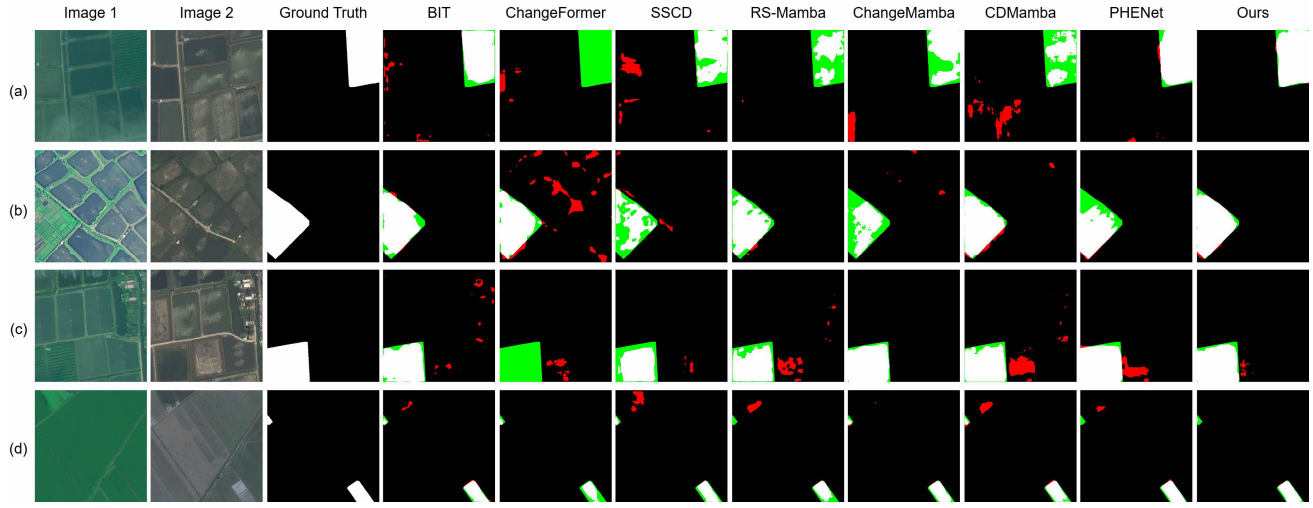
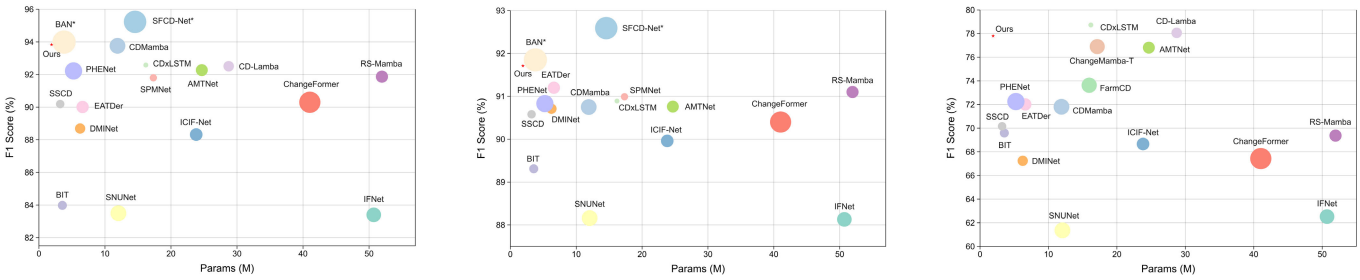


Fig. 7. Visualization results for qualitative comparison on the CLCD dataset. (a)–(d) Representative samples. Colors denote detection outcomes: true positives (white), true negatives (black), false positives (red), and false negatives (green).

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(a) (b) (c)

Fig. 8. Comparison of model parameters and $F1$-scores on three benchmark datasets. The radius of each point represents the logarithmic scale of FLOPs, where a larger radius implies greater computational complexity. "*" denotes the count of trainable parameters only. (a) WHU-CD dataset. (b) LEVIR-CD dataset. (c) CLCD dataset.

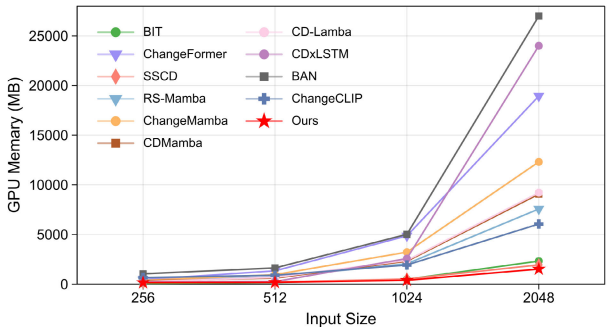


Fig. 9. Comparison of GPU memory usage during inference under various input sizes.

summarizes the tradeoffs across WHU-CD, LEVIR-CD, and CLCD. As illustrated in Fig. 8, the comparison of model parameters, FLOPs, and $F1$-scores across three benchmark datasets is presented. Fig. 8(a) and (b) shows that our method contains only 1.90M parameters and 1.84G FLOPs, achieving competitive performance that surpasses most representative methods. It is worth noting that the parameter counts of VFM-based methods shown in the figure include only trainable parameters, while their actual total parameter numbers are considerably higher. As shown in Fig. 8(c), our method ranks second only to CDxLSTM and CD-Lamba in terms of accuracy, yet requires substantially fewer parameters, demonstrating the high efficiency of TALS-CD.

Fig. 9 further presents the peak GPU memory usage of different methods during inference under various input resolutions. It can be observed that our model consistently requires less GPU memory than most other methods, particularly under large input sizes, highlighting its advantages in computational and memory efficiency.

D. Ablation Studies

In this section, we conduct a series of ablation experiments on three benchmark datasets, including WHU-CD, LEVIR-CD, and CLCD, to systematically evaluate the effectiveness of each key component and hyperparameter in the proposed method, as well as their impact on overall performance.

1) Ablation on TADE: To thoroughly evaluate the effectiveness of the proposed TADE module and its internal components, we conduct a set of ablation experiments on three benchmark datasets, namely, WHU-CD, LEVIR-CD, and CLCD. The quantitative results are summarized in Table III.

Variant A removes the entire TADE module and directly relies on the three CNN-based difference experts for bitemporal representation learning. As shown in Table III, this variant consistently yields inferior performance across all datasets and evaluation metrics. The performance degradation is particularly evident on CLCD, which contains complex scenes and subtle changes. These results demonstrate that TADE plays a critical role in modeling global difference dependencies and providing effective top-down guidance, which cannot be compensated for by local difference extraction alone.

To isolate the contribution of the TALS, Variant B retains the TADE structure but removes the TALS-enhanced branch, leaving only the main branch with a fixed scan order. Compared with Variant A, Variant B achieves noticeable performance improvements, confirming the effectiveness of TADE even without task-adaptive scanning. However, the performance gains remain limited, indicating that fixed scan strategies are insufficient to fully capture task-specific causal dependencies in CD.

Variant C further explores whether the explicit TALS-enhanced branch can be removed by directly replacing the fixed scan in the main branch with the TALS strategy. This modification leads to moderate improvements over Variant B, suggesting that task-aware scanning can indeed enhance global difference modeling. Nevertheless, Variant C still underperforms the full model. We attribute this behavior to the optimization dynamics of learnable scan paths. In the early stages of training, the task-aware scan priorities are not yet well established, resulting in unstable or suboptimal scan orders. When TALS is directly applied to the main branch, this instability may propagate through the entire data flow and negatively affect subsequent optimization, ultimately limiting performance.

In contrast, the full model (Ours) adopts a hybrid design, in which the main branch uses a fixed scan order to ensure stable sequence modeling, while a TALS-enhanced branch learns task-aware scan priorities to modulate the main features. This complementary design achieves the best performance across all datasets, as the fixed scan provides implicit regularization against early stage training instability, whereas the TALS

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TABLE III
ABLATION STUDY OF TADE AND ITS COMPONENTS ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

Variant	TADE	TALS Branch	Main Scan	WHU-CD			LEVIR-CD			CLCD		
				F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
A	×	×	N/A	91.94	85.08	99.30	91.34	84.05	99.12	75.89	61.14	96.36
B	✓	×	Fixed	92.48	86.02	99.36	91.54	84.39	99.15	76.28	61.65	96.54
C	✓	×	TALS	92.74	86.47	99.39	91.59	84.49	99.16	76.69	62.19	96.76
Ours	✓	✓	Fixed	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	**77.80**	**63.67**	**96.88**

TABLE IV
ABLATION STUDY OF MODE AND ITS COMPONENTS ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

Variant	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
only D_1	92.36	85.81	99.35	91.53	84.38	99.15	75.77	60.99	96.52
only D_2	92.87	86.69	99.40	91.54	84.40	99.15	75.63	60.81	96.36
only D_3	92.23	85.58	99.34	91.47	84.29	99.14	74.89	59.85	96.37
w/o Router	93.25	87.35	99.43	91.66	84.60	99.16	77.43	63.17	96.64
Full Model	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	**77.80**	**63.67**	**96.88**

TABLE V
EFFECT OF TALS SCAN DIRECTION ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

Scan Direction	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
Forward	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	**77.80**	**63.67**	**96.88**
Backward	93.68	88.12	99.46	91.67	84.62	99.16	77.59	63.39	96.74
Bidirectional	93.39	87.61	99.43	91.57	84.46	99.16	77.19	62.85	96.65

branch injects task-adaptive cues to emphasize change-relevant regions, leading to more reliable optimization and improved generalization.

2) Ablation on MoDE: Ablation results for the three experts (D_1–D_3) and the router are summarized in Table IV. Any single expert underperforms, indicating limited coverage of change cues. Using all three without routing improves $F1$ but still lags, since static fusion cannot exploit sample-dependent complementarity. Adding the router yields additional $F1$ gains of + 0.58, + 0.05, and + 0.37 on WHU-CD, LEVIR-CD, and CLCD, respectively. With all experts and routing enabled, MoDE adaptively fuses pixel-, channel-, and spatial-level differences and achieves the best performance, with similar trends for IoU and OA.

3) Effect of TALS Scan Direction: We further investigate the impact of scan direction in the TALS module, with results reported in Table V. Forward scanning consistently yields the best performance across all three datasets, while backward scanning shows comparable but slightly inferior results, and bidirectional scanning leads to a noticeable performance drop.

This degradation can be attributed to the incompatibility between bidirectional scanning and the learned task-aware scan priorities. While TALS propagates global context through a single causal pass with a consistent priority order, bidirectional scanning introduces a reverse traversal that interferes with this order, potentially delaying high-priority regions and disrupting causal continuity. As a result, task-aware difference aggregation becomes less effective, leading to inferior optimization behavior.

To further analyze the behavior of TALS and the difference between forward and backward scanning, we visualize the task-aware activation map and the corresponding scan paths in Fig. 10. Under the auxiliary supervision of the task-aware loss, TALS assigns higher priorities to changed regions and lower priorities to unchanged ones. Interestingly, the activation map reveals that boundary regions between change and nonchange areas are assigned the lowest priority, appearing in deeper blue. As a result, forward scanning models changed regions first, followed by background areas, while change boundaries are processed last. This ordering is well aligned with the dynamics of Mamba, where later tokens preserve more refined and stable states, thereby enabling more precise modeling of subtle-change boundaries compared to backward scanning.

4) Effect of Task-Aware Loss and Its Weight: We further analyze the role of the task-aware loss and its weighting strategy through two complementary ablation studies. As shown in Table VI, removing the task-aware loss leads to consistent performance degradation across all datasets, demonstrating that explicit task-aware supervision is essential for effective priority learning in TALS.

As shown in Table VII, we evaluate the sensitivity to the task-aware loss weight α. A moderate value ($\alpha = 0.1$) achieves the best performance, while both smaller and larger settings

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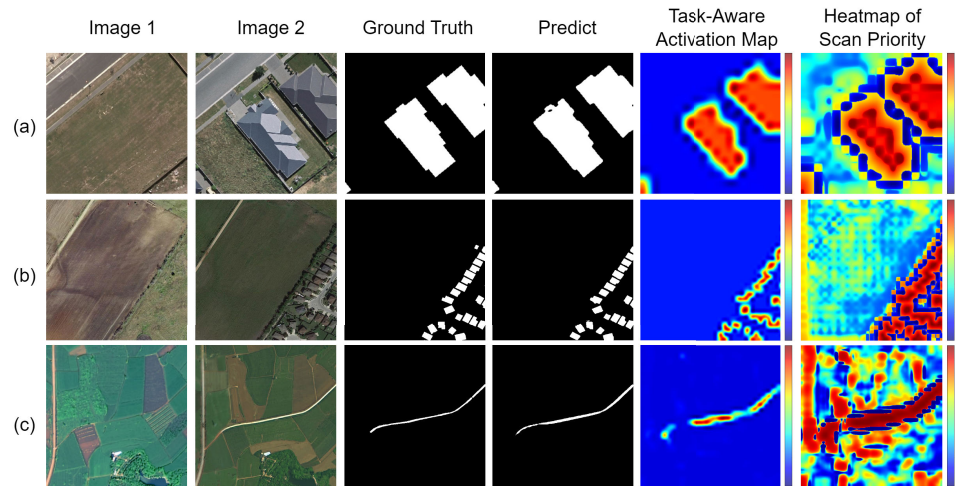


Fig. 10. Visualization of task-aware activation maps and the corresponding scan-priority heatmaps. Warmer (redder) colors indicate higher weights or scanning priorities. (a) WHU-CD dataset. (b) LEVIR-CD dataset. (c) CLCD dataset.

TABLE VI

ABLATION STUDY OF THE TASK-AWARE LOSS ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

Variant	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
w/o Task-Aware Loss	92.64	86.30	99.37	91.31	84.01	99.13	75.09	60.11	96.26
Full Loss	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	**77.80**	**63.67**	**96.88**

TABLE VII

EFFECT OF THE TASK-AWARE LOSS WEIGHT α ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

α	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
0.01	93.25	87.35	99.42	91.39	84.15	99.14	76.63	62.12	96.59
0.1	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	**77.80**	**63.67**	**96.88**
1	93.59	87.95	99.45	91.63	84.55	99.15	77.11	62.74	96.74
10	93.05	87.00	99.40	91.52	84.37	99.15	76.36	61.76	96.63
100	92.71	86.42	99.38	91.49	84.32	99.14	76.11	61.43	96.60

TABLE VIII

EFFECT OF LOCAL SCANNING WINDOW SIZE IN TADE ON THREE BENCHMARK DATASETS. BEST RESULTS ARE IN BOLD

Window Size	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
(1, 1, 1)	93.00	86.91	99.41	91.67	84.62	99.16	77.01	62.61	96.75
(2, 2, 2)	93.37	87.56	99.43	91.69	84.65	99.16	77.53	63.30	96.79
(4, 4, 4)	93.09	87.07	99.41	91.68	84.64	99.16	77.66	63.48	96.83
(1, 2, 4)	**93.83**	**88.38**	**99.47**	**91.71**	**84.68**	**99.16**	77.59	63.39	96.74
(2, 4, 8)	93.45	87.71	99.44	91.60	84.51	99.16	**77.80**	**63.67**	**96.88**
(4, 8, 16)	93.09	87.07	99.41	91.47	84.27	99.14	76.53	61.98	96.74

lead to degradation. When α is too small, the supervision on the TALS branch is insufficient, resulting in slow and ineffective learning of task-aware scan priorities and thus limiting its causal modeling capability. In contrast, an overly large α causes the task-aware loss to dominate the optimization, interfering with the learning of the main CD objective and destabilizing the training process. These results indicate that the task-aware loss should act as an auxiliary supervisory signal that balances effective causal guidance with stable optimization of the primary task.

5) Effect of Local Scanning Window Size in TADE: We investigate the influence of local scanning window sizes adopted in the three TADE modules, as summarized in Table VIII. We evaluate both single-scale configurations, where all TADE modules share the same window size, and multiscale configurations with progressively increasing

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window sizes. In particular, the $(1, 2, 4)$ configuration achieves the best performance on WHU-CD and LEVIR-CD, whereas $(2, 4, 8)$ performs best on CLCD. This variation reflects the differing spatial characteristics of the datasets: WHU-CD and LEVIR-CD mainly involve compact building changes that benefit from finer local scanning, while CLCD contains more heterogeneous and larger scale land-cover changes, where moderately larger receptive fields become advantageous. These results suggest that combining different window sizes allows TADE to adapt to diverse change patterns rather than relying on a single fixed scale. Moreover, we observe that introducing multiscale configurations on top of single-scale settings generally improves performance, as different window sizes capture complementary change patterns. However, excessively large window sizes may degrade performance due to the loss of fine-grained spatial details. This suggests that a proper balance between local detail preservation and global context modeling is crucial for achieving optimal performance.

V. Conclusion

This article presented TALS-CD: a causal continuity-based task-aware Mamba (TALS) for RSCD. TALS learns task-aware scan paths for causal continuity modeling; coupled with TADE for global difference guidance and MoDE routing for top-down refinement, it produces compact yet accurate change maps. On WHU-CD, LEVIR-CD, and CLCD, the model achieves strong results under low parameter and FLOP budgets, and ablations validate the contribution of each component. An immediate extension is to endow TALS with adaptive local windowing, enabling data-driven window selection per region and stage. However, TALS still relies on task-specific supervision and introduces some architectural complexity, which may limit its generalization to more complex scenarios. Future work will focus on more adaptive and efficient designs.

References

[1] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[2] H. Chen, E. Nemni, S. Vallecorsa, X. Li, C. Wu, and L. Bromley, "Dual-tasks Siamese transformer framework for building damage assessment," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1600–1603.

[3] X. Li, F. Ling, G. M. Foody, and Y. Du, "A superresolution land-cover change detection method using remotely sensed images with different spatial resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 3822–3841, Jul. 2016.

[4] R. E. Kennedy et al., "Remote sensing change detection tools for natural resource managers: Understanding concepts and tradeoffs in the design of landscape monitoring projects," *Remote Sens. Environ.*, vol. 113, no. 7, pp. 1382–1396, Jul. 2009.

[5] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," 2024, *arXiv:2401.09417*.

[6] J. Jiao et al., "VMamba: Visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 103031–103063.

[7] T. Huang, X. Pei, S. You, F. Wang, Q. Chen, and C. Xu, "LocalMamba: Visual state space model with windowed selective scan," in *Proc. Eur. Conf. Comput. Vis.*, 2024, pp. 12–22.

[8] B. Li, H. Zhao, W. Wang, P. Hu, Y. Gou, and X. Peng, "MaIR: A locality-and continuity-preserving mamba for image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 7491–7501.

[9] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[10] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[11] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[12] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[13] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[14] Z. Sun, H. Song, K. Zhang, G. Dong, L. Liang, and Y. Zhao, "Segment anything model guided semantic knowledge learning for remote sensing change detection," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 5830–5834.

[15] C. Yang et al., "PlainMamba: Improving non-hierarchical mamba in visual recognition," 2024, *arXiv:2403.17695*.

[16] Y. Xiao et al., "GrootVL: Tree topology is all you need in state space model," 2024, *arXiv:2406.02395*.

[17] C. Ma, Z. Wang, F. Xie, and W. Zhang, "QuadMamba: Learning quadtree-based selective scan for visual state space model," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 117682–117707.

[18] Z. Wu et al., "CD-Lamba: Boosting remote sensing change detection via a cross-temporal locally adaptive state space model," 2025, *arXiv:2501.15455*.

[19] L. Liu et al., "DefMamba: Deformable visual state space model," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 8838–8847.

[20] H. Wei et al., "Multiscale spatial frequency fusion and prior change guidance network for remote sensing change detection," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–13, 2025.

[21] X. Zuo et al., "Integrating segment anything model with instance-level change generation for single-temporal unsupervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631717.

[22] C. Tao, D. Kuang, Z. Huang, C. Peng, and H. Li, "HASNet: A foreground association-driven Siamese network with hard sample optimization for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614416.

[23] Y. Zheng, J. Zhao, and J. Qu, "A spatial–spectral difference guided DDPM for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5527714.

[24] J. Wang et al., "SPMNet: A Siamese pyramid mamba network for very-high-resolution remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4410314.

[25] G. Wang, H. Chen, T. Qiao, J. Wang, and W. Liu, "Resolution-difference embedded network for cross-resolution remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5637721.

[26] J. Li, F. Shao, X. Meng, and Z. Yang, "Dual-branch cross-resolution interaction learning network for change detection at different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4600116.

[27] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[28] Y. Xu et al., "From macro to micro: A lightweight interleaved network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4406114.

[29] W. Yang, Z. Ye, L. Mei, Y. Yao, J. Chen, and Y. Li, "DDRL: Domain distribution reconstruction learning for binary change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5604713.

[30] C. Liu, K. Chen, H. Zhang, Z. Qi, Z. Zou, and Z. Shi, "Change-agent: Toward interactive comprehensive remote sensing change interpretation and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635616.

[31] C. Han, C. Wu, M. Hu, J. Li, and H. Chen, "C2F-SemiCD: A coarse-to-fine semi-supervised change detection method based on consistency regularization in high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702621.

[32] Z. Zhang, L. Bao, S. Xiang, G. Xie, and R. Gao, "B2CNet: A progressive change boundary-to-center refinement network for multitemporal remote sensing images change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11322–11338, 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 03:23:24 UTC from IEEE Xplore. Restrictions apply.

[33] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[34] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[35] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[36] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[37] R. Zhang, H. Zhang, X. Ning, X. Huang, J. Wang, and W. Cui, "Global-aware Siamese network for change detection on remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 199, pp. 61–72, May 2023.

[38] W. Liu, Y. Lin, W. Liu, Y. Yu, and J. Li, "An attention-based multiscale transformer network for remote sensing image change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 202, pp. 599–609, Aug. 2023.

[39] J. Ma, J. Duan, X. Tang, X. Zhang, and L. Jiao, "EATDer: Edge-assisted adaptive transformer detector for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602015.

[40] L. Wang, Y. Fang, Z. Li, C. Wu, M. Xu, and M. Shao, "Summator–subtractor network: Modeling spatial and channel differences for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5604212.

[41] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[42] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[43] D. Zhang, F. Wang, L. Ning, Z. Zhao, J. Gao, and X. Li, "Integrating SAM with feature interaction for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4513011.

[44] S. Dong, L. Wang, B. Du, and X. Meng, "ChangeCLIP: Remote sensing change detection with multimodal vision-language representation learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 53–69, Feb. 2024.

[45] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314.

[46] Z. Wu, X. Ma, R. Lian, K. Zheng, and W. Zhang, "CDxLSTM: Boosting remote sensing change detection with extended long short-term memory," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[47] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[48] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[49] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[50] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[51] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[52] J. Yuan, S. Chen, J. Zhou, and L. Zhao, "PHENet: Toward robust fog-resilient change detection in remote sensing imagery with a physics-guided height-enhanced network," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 3000611.

[53] H. Wang, N. Wang, and X. Li, "FarmCD: Agricultural information-guided gated network for farmland change detection from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 4401714.

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